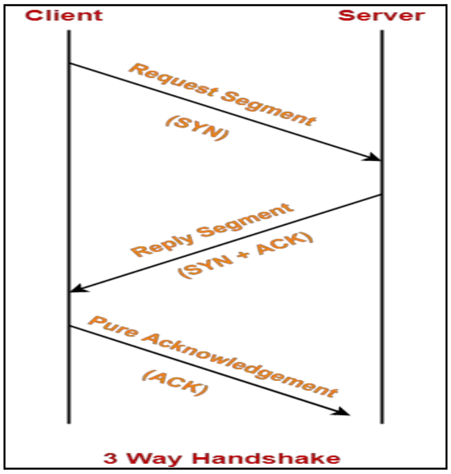
Question 1:

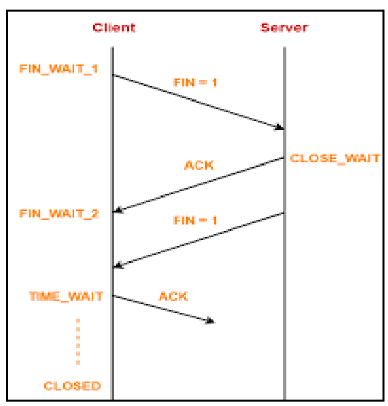
a) Explain the mechanism of connection establishment [1 mark]:[0.5 figure + 0.5 correct explanation]

3-way handshake



Explain the mechanism of the connection termination [1 mark]:[0.5 figure + 0.5 correct explanation]

1. 2-way termination [Because its full duplex mode][following figure consider it both]



b) 1. Reason for the need of handshaking algo for connection establishment [1 mark]

Why? In 3-way handshake, it may be possible that the client/sender thinks that the connection is aborted.and the server/receiver thinks that the connection is still there.And due to that reason,resources are continuosly consumed by server/receeiver which needs to be free.

2. Reason for the need of handshaking algo. for connection termination [1 mark]

While client/sender have completed sending the packets there is still a possibility that data from server/receiver is till present in the pipline.So A must wait for a FINISH flag from receiver side to close the connectio its side so that is why it is also known as half closure.
Connection closes only after both half closure is completed.

Question 2:
Marks:
1 = Full hop count routing algorithm
1 = Structure of Routing table and Link State Packet
a)
Routing table generation:
Collecting neighbourhood link information
Send hello packet on all links, get responses from the connected neighbours.
Control flooding the network with “Link State Packet” so that it reaches all connected nodes.
Combine all LSPs to generate weighted graph of the network.
Run Dijkstra’s algorithm to find the shortest path.
Generate routing table that has entries for each destination.

Routing table structure:

Destination	Next hop

LSP Structure:

Seq No.	Src=A	B	hop=1	C	hop=1	D	hop=1

b)
Marks:
1 = stable routing table for hop count as the matric.
1 = unstable routing table for delay as the matric.
Hop count is static. So the values of the routing table remains same for each link.
So the routing table remains stable, which is desirable.
Delay will change as the traffic changes on each link due to queuing delay, transmission delay etc.

As the routing table changes, more traffic moves on the shortest path (according to delay) Which in turn increases the delay on that path, so the shortest path changes and the routing table needs to be rearranged.

This leads to oscillating (unstable) routing table. Which is not desirable.

Question 3 Solution:

Q-3

Estimated RTT = (1 − α) Estimated RTT + α Sample RTT

a] $\boxed{\text{Timeout Interval} = \text{Estimated RTT} + 4 * \text{DevRTT}}$

DevRTT = (1 − β) * DevRTT + β |Sample RTT − Estimated RTT|

1 mark for

b] $\alpha = 1/4$

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b] $\alpha = 1/4$

$$\text{Estimated RTT}^{n+1} = (1-\alpha)\text{Estimated RTT}^{n} + \alpha * \text{SampleRTT}^{n}$$

$$= (1-\alpha)\left[(1-\alpha)\text{Estimated RTT}^{n-1} + \alpha * \text{sample RTT}^{n-1}\right] + \alpha * \text{Sample RTT}^{n}$$

$$= \left(1-\frac{1}{4}\right)\left[\left(1-\frac{1}{4}\right)\text{ERTT}^{n-1} + \frac{1}{4}\text{Sample RTT}^{n-1}\right] + \frac{1}{4} * \text{Sample RTT}^{n}$$

$$= \frac{3}{4}\left[\frac{3}{4}\text{ERTT}^{n-1} + \frac{1}{4}\text{Sample RTT}^{n-1}\right] + \frac{1}{4}\text{sample RTT}^{n}$$

$$= \frac{9}{16}\text{ERTT}^{n-1} + \boxed{\frac{3}{16}}\text{sample RTT}^{n-1} + \boxed{\frac{1}{4}}\text{Sample RTT}^{n}$$

Correct derivation of equation 1½ marks ½ marks for correct weight

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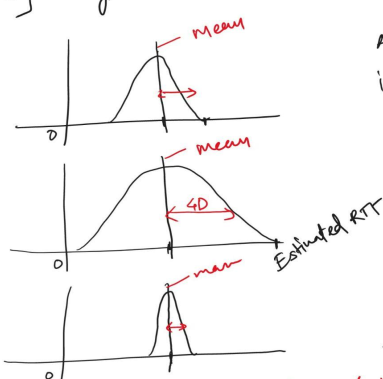
c] Why to use both RTT mean & variance ?



All the distribution plots shown in left side are distribution of RTT and mean is same for all of them but because of variance in RTT the width of the distribution varies, if we only consider mean value then estimation might be too small or too high and not exact.

Correct explaination → 1 mark

Question 4:
Answer:

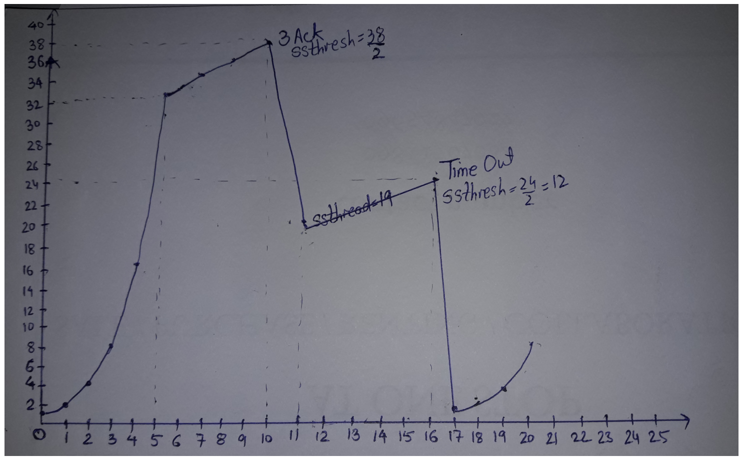
a)
Flow Control: matching transmitter and receiver data rates. (0.5 marks)
Congestion Control: managing the traffic through the routers to control queuing delays and packet loss. (0.5 marks)

b)
Slow Start Phase : exponential increment – In this phase after every RTT the congestion window size increments exponentially.

Congestion Avoidance Phase : additive increment – This phase starts after the threshold value also denoted as ssthresh. The size of cwnd(congestion window) increases additive. After each RTT cwnd = cwnd + 1.

Congestion Detection Phase : multiplicative decrement
Case 1 : Retransmission due to Timeout – In this case congestion possibility is high.
(a) ssthresh is reduced to half of the current window size.
(b) set cwnd = 1
(c) start with slow start phase again.

Case 2 : Retransmission due to 3 Acknowledgement Duplicates – In this case congestion possibility is less.
(a) ssthresh value reduces to half of the current window size.
(b) set cwnd= ssthresh
(c) start with congestion avoidance phase



c)

Time out implies congestion.

Possible reasons for time out are:

1. Packet corrupted.
2. Mis routing.
3. Packet loss at the router (congention).

Probability of 1 and 2nd is very low compared to congestion. It means Time out implies congestion in most of the time.

d) Rate (window size) increase in every RTT. if RTT is low, window(rate) increase faster and hence higher throughput.